Parsimonious and Explainable Fraud Detection: A Hybrid Evolutionary Approach to Mitigate Circularity in XAI

Ederlei Rodrigo dos Reis[1][https://orcid.org/0000−0002−4740−2483], Alessandra Aparecida Paulino[1][https://orcid.org/0000−0003−2249−2977], and Marcelo Keese Albertini[1][https://orcid.org/0000−0002−1846−946X]

[1] Universidade Federal de Uberlândia, Uberlândia, Minas Gerais, Brazil
{ederleireis, alessandra, albertini}@ufu.br

Abstract. Credit card fraud detection requires models that are not only highly accurate but also auditably transparent. However, the extreme class imbalance and non-linear interactions between features often lead to complex "black-box" solutions. This study presents a hybrid evolutionary framework (GA-Wrapper) that integrates a Genetic Algorithm with a cost-sensitive LightGBM to optimize the trade-off between predictive power and parsimony. Our results show that the proposed method identifies a "vital core" of 7 features, achieving a Gini coefficient of 0.8032 and an AP (Average Precision) of 0.7126, significantly outperforming greedy methods like RFE. Furthermore, we validate the solution using XAI 2.0 metrics, achieving a 51.72% increase in compactness and 100% axiomatic fidelity via TreeSHAP. The findings demonstrate that global evolutionary optimization can effectively reduce cognitive load for human auditors while maintaining high operational recall (0.8176), providing a robust path for Trustworthy AI in financial services.

Keywords: Credit Card Fraud Detection · Genetic Algorithms · Feature Selection · Explainable AI (XAI) · Trustworthy AI · LightGBM · SHAP · Parsimony.

1 Introduction

Fraud detection is a critical challenge at the intersection of Data Science and Cybersecurity [23], where losses extend beyond financial values to the erosion of consumer trust [13]. These domain is characterized by extreme class imbalance—where fraudulent transactions often constitute less than 0.1% of the total volume—and by an adversarial dynamic, where attack patterns continuously evolve to evade pre-established static rules.

The domain is characterized by extreme class imbalance and adversarial dynamics, leading the industry to migrate from rule-based systems to high-capacity ensembles like LightGBM. However, these "black-box" models suffer from structural opacity, hindering explanations for regulators and auditors [1].

With the enactment of global regulations such as the GDPR in Europe and the LGPD in Brazil, the "right to explanation" has become a legal requirement, transforming explainability from a competitive differentiator into a compliance obligation. Legal certainty depends not only on technical accuracy, but also on the ability to adapt the explanation to the cognitive limitations of specific users (auditors versus end clients), an engineering challenge that massive models cannot overcome [15].

In response, the field of Explainable Artificial Intelligence (XAI) has experienced exponential growth. Molnar et al. [31] describe the historical evolution of XAI, noting that while roots exist since classical regression, the recent explosion focuses on model-agnostic methods. While methods like SHAP [29] emerged as gold standards, the first generation of XAI reached a plateau by neglecting the human recipient [28]. Mathematically correct explanations with excessive variables are cognitively useless under pressure, highlighting a disconnect between optimization and utility [42].

Furthermore, we identify a methodological flaw in recent literature: incorporating explainability metrics directly into the loss function [11]. This introduces a "vicious circularity" where explanations become artifacts manufactured to satisfy constraints rather than independent audits, a manifestation of Goodhart's Law [16]. Such over-optimization can hide complex dependencies, violating the principle of fidelity [22].

We hypothesize that true explainability in XAI 2.0 must emerge from attribute reduction and predictive competence. We propose using Genetic Algorithms (GA) to identify "vital players" [32] that maximize performance while minimizing structural complexity. The main contributions are:

1. An evolutionary fitness function based on Occam's Razor and Pareto efficiency [36].
2. A benchmark demonstrating GA's superiority in finding "blind synergies" in non-monotonic data [38].
3. Integration of Fidelity and Compactness metrics with Cognitive Load Theory [28].

2 Related Work

This section contextualizes our proposal within Guidotti's and Longo's taxonomies, highlighting the gaps in current explainability and feature selection strategies.

2.1 Taxonomy and Surrogates: Local vs. Global

Guidotti et al. [19] distinguish between local and global explainability. Dominant strategies often employ surrogate models, such as LORE [20], which uses GAs for local rule extraction, or symbolic global surrogates via Genetic Programming [14]. However, these intermediaries face fidelity issues [22], as the surrogate may fail to reflect the original model's logic accurately. Our approach eliminates this layer

by simplifying the original LightGBM structure through global feature selection, leveraging TreeSHAP [30] to guarantee 100% axiomatic fidelity at the structural root.

2.2 Feature Selection Philosophies

Trait selection is divided into two main philosophies:

1. **All-Relevant:** Methods like Boruta [26] retain all correlated variables. While robust, they generate dense models that fail the XAI 2.0 compactness criterion.
2. **Minimal-Optimal:** Lasso (L_1) and RFE seek minimal sets but often fail in complex fraud domains; Lasso assumes linearity, while RFE is a greedy method assuming monotonicity, missing the non-linear synergies described by Siedlecki and Sklansky [38].

Our GA proposal navigates this non-monotonic space to find the "Minimal-Optimal" subset that RFE/Lasso miss, with the parsimony Boruta ignores.

2.3 Frontiers of Financial and Legal XAI

Recent advances published in the Communications in Computer and Information Science series on Explainable Artificial Intelligence in 2025 highlight the urgency of new approaches. Van Veen et al. [41] propose a causal perspective for fraud detection, arguing that traditional methods do not distinguish correlation from causation. While we recognize the importance of formal causal inference, our approach offers a pragmatic alternative focused on "robustness": GA seeks to discard noisy variables and maintain consistent predictors.

Nguyen et al. [33] presented an approach based on semi-supervised Graph Neural Networks (GNNs) with Granger-Causal explanations. While powerful for capturing network topology, this approach adds a significant layer of structural (the GNN) and temporal complexity. In contrast, our proposal for "evolutionary parsimony" in tabular data offers a computationally more efficient and cognitively more accessible solution as a "grey-box" for auditors who are not graph experts.

Giudici et al. [18] addressed cyber risk using time-varying models. We argue that highly parameterized models are more susceptible to temporal degradation (model drift). A parsimonious model, focused on a few robust causal attributes selected by our GA, offers greater temporal stability, facilitating ongoing risk management.

Finally, Fairhurst et al. [15] mapped the multidisciplinary legal requirements for explainability, emphasizing that "trust" depends on adapting the explanation to the specific user. Our methodology directly addresses the engineering gap they identified: by drastically reducing dimensionality (maximum number of attributes), we make the explanation feasible for the working memory of a human auditor, meeting the requirements of User-Specific Explainability without the need for complex additional interfaces. Babaei et al. [2] discuss explainable fairness

in lending; although our focus is on fraud detection, the simplicity of our model facilitates bias auditing by removing discriminatory proxies that could hide in dense models.

2.4 Sparsity, Actionability, and the Human-in-the-Loop

Manual SHAP-guided selection [46] lacks scalability for modern transaction volumes. By automating selection through GA, we replicate human intelligence without cognitive fatigue. This reduction also trivializes the generation of counterfactuals; as defined by Guidotti [21], effective counterfactuals require sparsity and actionability. In our low-dimensional space, identifying necessary changes for classification reversal becomes computationally efficient and semantically straightforward.

3 Theoretical Foundation

Our framework integrates human-centric XAI 2.0, the epistemology of simplicity, evolutionary computation, and game theory to create robust fraud models.

3.1 XAI 2.0 and Cognitive Load Management

XAI 2.0 shifts focus from algorithm-centric to human-centric explanations [28]. Cognitive Load Theory indicates that humans efficiently process only 7 ± 2 elements (Miller's famous magic number); exceeding this leads to "explanation fatigue" [28].

Vilone e Longo [43] propose objective metrics for evaluating explanations, highlighting "compactness" and "fidelity" as orthogonal dimensions of quality.

- Compactness: Refers to the sparsity of the explanation. The fewer elements (features), the lower the extrinsic cognitive load.
- Fidelity: Refers to how accurately the explanation reflects the behavior of the underlying model.

Our approach uses GA as an automated cognitive filter, removing variables that contribute to noise (increasing extrinsic cognitive load) without contributing to the signal (Gini coefficient). This transforms the explanation from a dense technical artifact into an accessible cognitive tool. Longo et al. [27] identify the lack of unified evaluation metrics as a key challenge, and our proposal to use evolution-induced parsimony directly addresses this challenge.

3.2 Occam-PAC Dialectics: Navigating Simplicity

Applying Occam's Razor to PAC-Learning (Probably Approximately Correct), Blumer et al. [6] proved the following theorem about hypothesis size: if a learning algorithm produces a hypothesis h consistent with m examples of training, and the complexity de h is significantly smaller than m, so h has a guaranteed upper

bound for the generalization error ϵ. Mathematically, for a space of hypotheses H, the number of samples m necessary to ensure a smaller error than ϵ with probability $1 - \delta$ is:

$$m \geq \frac{1}{\epsilon}\left(\ln|H| + \ln\frac{1}{\delta}\right) \tag{1}$$

By reducing the number of features via GA, we are effectively reducing $|H|$, the size of the space of possible hypotheses, which, according to Blumer, increases the theoretical probability of generalization to unseen data.

However, excessive simplicity can lead to underfitting in high-entropy domains [12] or slow learning rates [4]. We navigate this by pairing a high-capacity LightGBM classifier with GA-driven dimensionality reduction to ensure comprehensibility [17].

3.3 Evolutionary Search in Non-Monotonic Spaces

Feature selection is an NP-Hard combinatorial problem (2^d space). Traditional greedy methods fail when feature sets do not follow monotonicity [38]. GAs are ideal for navigating these non-monotonic spaces, as population-based searches explore synergistic combinations discarded by greedy heuristics [38, 45]. Binary representation is superior for noise elimination in real-time systems [3, 24]. Our approach uses GA for structural optimization [35, 47], employing Pareto scalarization [36] to balance accuracy and parsimony [25, 44].

3.4 Game Theory and Vital Players

We utilize the Shapley Value [34, 37]—which satisfies the axioms of Efficiency, Symmetry, Null Player, and Additivity—to validate interpretability. Specifically, TreeSHAP [30] provides exact, polynomial-time calculations ($O(TLD^2)$) suitable for high-risk binary classification [5]. We define a "vital player" i in the cooperative game (N, v) as one whose removal converts a winning coalition into a losing one [32]:

$$v(S) = 1 \quad \text{e} \quad v(S \setminus \{i\}) = 0 \tag{2}$$

The GA identifies this minimal subset of vital attributes while discarding null players that pollute explanations [32].

3.5 The Gini Coefficient as an Objective Function

We employ the Inequality Gini Coefficient ($2 \cdot AUC - 1$) rather than Gini Impurity [40]. Gini is superior to information gain for sparse, long-tailed data [39] and serves as a mathematically stable objective function for guiding evolutionary algorithms in complex stochastic optimization [9].

4 Methodology

Our framework is a hybrid wrapper method where a Genetic Algorithm (GA) optimizes the input structure for a LightGBM inductor. The architecture is divided into three functional modules.

4.1 Module 1: Preprocessing and Robustness

Financial fraud data are characterized by long-tailed distributions and severe outliers (e.g., legitimate transactions of atypical value). Standard normalization (Z-score) is sensitive to these extremes, distorting the scale for most data. We adopted the Robust Scaler, which uses order statistics (Median and Interquartile Range - IQR):

$$x_{scaled} = \frac{x - Q_2(x)}{Q_3(x) - Q_1(x)} \tag{3}$$

A stratified hold-out strategy is employed, isolating a test set (X_{test}) in a "vault" to ensure an unbiased post-evolutionary evaluation.

4.2 Module 2: The Evolutionary Engine

GA was designed to operate in the binary search space $\{0, 1\}^d$, where d is the total number of attributes. In our execution, each individual is a bit vector (chromosome) of length d. The gene $i = 1$ indicates the presence of the attribute, and $i = 0$ his absence. The initial population is randomly generated using a Bernoulli distribution ($p = 0.5$), ensuring initial diversity in the search space.

Fitness Function: The fitness function is the core of our parsimony proposal. It seeks not only accuracy, but also the economic balance of information, approximating a solution on the Pareto front [36] through scalarization. We define fitness $F(x)$ as:

$$F(\mathbf{x}) = \begin{cases} -1.0 & \text{if } d(\mathbf{x}) = 0 \\ \bar{g}(\mathbf{x}) - \lambda \cdot \left(\frac{d(\mathbf{x})}{D}\right) & \text{otherwise} \end{cases} \tag{4}$$

Where:

- $\mathbf{x}$: The binary feature selection vector, where $x_j = 1$ denotes the inclusion of the j-th feature and $x_j = 0$ its exclusion.
- $\bar{g}(\mathbf{x})$: The mean Gini coefficient obtained via stratified 3-fold cross-validation on the training set. This metric serves as a robust estimator of predictive power, minimizing the variance of the assessment.
- λ: The parsimony penalty factor. This hyperparameter governs the aggressiveness of the regularization (Occam's Razor). A higher λ enforces the GA to prune attributes that offer only marginal performance gains (Performance Plateau), strictly adhering to the *Compactness* criterion defined by Longo et al. [27].

– $d(\mathbf{x})$: The cardinality of the selected subset (effective dimensionality), defined as $d(\mathbf{x}) = \sum_{j=1}^{D} x_j$.

The algorithm 1 details the computational procedure.

Algorithm 1: Parsimonious Fitness Evaluation (Evolutionary Wrapper)

Input: Binary Chromosome $\mathbf{b}$, Dataset $\mathcal{D} = (\mathbf{X}, \mathbf{y})$, Penalty λ, Total Features D
Output: Fitness Score F
`// 1. Genotype Decoding`
$S \leftarrow \{i \mid \mathbf{b}[i] = 1\}$ `// Active indices`
$d' \leftarrow |S|$ `// Current dimensionality`
`// 2. Hard Penalty for Empty Set`
if $d' = 0$ **then**
  return -1.0
end
`// 3. K-Fold Cross-Validation Assessment`
$G \leftarrow \emptyset$
Split $\mathcal{D}$ into K stratified folds $\{(\mathbf{X}_k^{tr}, \mathbf{y}_k^{tr}), (\mathbf{X}_k^{val}, \mathbf{y}_k^{val})\}_{k=1}^{K}$
for $k \leftarrow 1$ **to** K **do**
  $M_k \leftarrow \text{TrainLightGBM}(\mathbf{X}_k^{tr}[:, S], \mathbf{y}_k^{tr})$
  $\hat{\mathbf{y}}_k \leftarrow \text{PredictProba}(M_k, \mathbf{X}_k^{val}[:, S])$
  $auc_k \leftarrow \text{AreaUnderCurve}(\mathbf{y}_k^{val}, \hat{\mathbf{y}}_k)$
  $gini_k \leftarrow 2 \cdot auc_k - 1$
  $G \leftarrow G \cup \{gini_k\}$
end
$\bar{g} \leftarrow \text{mean}(G)$
`// 4. Parsimony Regularization (Occam's Razor)`
$Cost \leftarrow \lambda \cdot \left(\frac{d'}{D}\right)$
$F \leftarrow \bar{g} - Cost$
return F

Genetic Operators:

– Selection: Tournament Selection with size $k = 3$. This method maintains selective pressure without losing diversity as quickly as "Roulette", preventing premature convergence.
– Crossover: Uniform. Unlike single-point crossover, uniform crossover treats each gene independently, which is ideal for feature selection where there is no spatial dependency between adjacent columns (unlike pixels in an image), as observed by Hussein et al. [24].
– Mutation: *Bit-flip* with low probability ($1/d$), allowing the reintroduction of discarded attributes or the removal of redundant attributes to escape from local optimum.

4.3 Module 3: Post-Hoc Validation via TreeSHAP

After convergence, the optimal subset is validated using TreeSHAP [1] [2] on X_{test}. The computational complexity of TreeSHAP is $O(TLD^2)$, where T is the number of trees, L the maximum depth, and D the number of leaves, making it efficient for large-scale validation. Comparing the summary plots of the optimized model versus the model with all features allows visualization of the reduction in cognitive load and the elimination of noise, validating the effectiveness of the approach in generating QII transparency [11] and meet the XAI quality criteria defined by Longo et al [28].

5 Experimental Setup

This section details the computational environment, data partitioning strategies, and the specific hyperparameters configured for the evolutionary engine, ensuring the reproducibility of the experiments. Code is available at the project repository[3].

5.1 Data Preparation and Partitioning

The empirical analysis utilizes a dataset[4] of credit card transactions conducted by European cardholders in September 2013. The data covers a two-day period, comprising 284,807 transactions, of which only 492 are classified as fraudulent, resulting in a highly imbalanced class distribution (0.172% positive class). Due to privacy constraints, the majority of input variables (V1 through V28) are the result of a Principal Component Analysis (PCA) transformation. Only two features remain in their original form: "Time", representing the seconds elapsed since the first transaction, and "Amount", denoting the transaction value. The target variable "Class" is binary, where 1 indicates fraud and 0 denotes legitimate activity [8, 10]. In summary, the dataset consists of 284,807 transactions with no missing values. The 'Time' feature was discarded as it does not contribute to the structural detection of fraud patterns in this snapshot context. To mitigate the impact of outliers in the 'Amount' variable, we applied the *RobustScaler* (based on the Interquartile Range), as defined in Eq. 3.

Given the extreme class imbalance (only 492 frauds, $\approx 0.17\%$), we employed a Stratified Hold-out strategy with a fixed random seed ($seed = 42$) to preserve the class distribution. The data was split as follows:

- **Training Set (Evolutionary Search):** 70% of the data ($N_{train} = 199,364$), containing 344 fraud instances. This subset is used exclusively by the GA to compute fitness.

[1] https://github.com/shap/shap/blob/master/shap/explainers/_tree.py
[2] https://shap.readthedocs.io/en/latest/generated/shap.TreeExplainer.html
[3] https://github.com/e-reis/hybrid_evolutionary_approach_to_fraud_detection
[4] https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

– **Test Set:** 30% of the data ($N_{test} = 85,443$), containing 148 fraud instances. This set remains strictly inaccessible during the optimization phase to prevent data leakage and is used only for the final post-hoc validation.

5.2 Evolutionary Engine Configuration

The Genetic Algorithm was implemented using the DEAP framework, configured to maximize the parsimonious fitness function defined in Eq. 4. The search space is represented by binary vectors of length $D = 29$ (28 PCA components + Amount). The evolutionary operators were tuned as follows:

– **Selection:** Tournament Selection (size = 3) to maintain selective pressure.
– **Crossover:** Uniform Crossover with probability $P_{cx} = 0.1$, promoting gene mixing without positional bias.
– **Mutation:** Bit-flip Mutation with probability $P_{mut} = 0.05$ (approx. $1/D$), allowing exploration of the search space boundaries.
– **Population & Generations:** A population of 50 individuals evolved over 50 generations.

5.3 Internal Fitness Evaluation (The Wrapper)

To evaluate each individual (subset of features) within the GA, we utilized a "lightweight" instance of the LightGBM classifier to balance computational cost and accuracy. The internal evaluation protocol includes:

– **Validation Strategy:** Stratified 3-Fold Cross-Validation on the Training Set to minimize variance in the Gini estimate ($\bar{g}$).
– **Internal Model Hyperparameters:** LightGBM trained with $n_estimators = 50$, $learning_rate = 0.1$, and $num_leaves = 31$.
– **Imbalance Handling:** Instead of computational expensive oversampling techniques (e.g., SMOTE), we utilized a cost-sensitive learning approach via the `scale_pos_weight` parameter. This weight was dynamically calculated for each training fold as ratio $w = N_{majority}/N_{minority}$. This ensures that the evolutionary search penalizes false negatives proportionally to the specific imbalance of the fold, aligning the fitness score with the calibrated probability space required for the final model.
– **Parsimony Pressure:** The penalty factor was set to $\lambda = 0.05$, forcing the algorithm to discard features that do not contribute significantly to the predictive gain.

6 Results and Performance Analysis

This section presents the empirical findings obtained from the experimental setup described in Section 5. We analyze the evolutionary trajectory of the optimization engine, the stability of the feature selection process, and the comparative performance against state-of-the-art methods.

It is crucial to emphasize that all results reported herein were computed strictly on the withheld test set (the "vault"), which comprises 30% of the original dataset ($N_{test} = 85,443$). This protocol ensures an unbiased evaluation, preventing any form of data leakage from the evolutionary search phase.

We emphasize that we chose to run the entire procedure ten times with the same inputs and scenarios, without any changes. And to represent the results shown here, we used the results from the tenth run.

6.1 Evolutionary Dynamics and Convergence Analysis

The optimization process was monitored over 50 generations to assess the trade-off between predictive accuracy (Gini) and model parsimony (Compactness). Figure 1 illustrates the convergence behavior of the Genetic Algorithm, tracking both the best individual and the population average.

The initial population, generated with a Bernoulli distribution ($p = 0.5$), started with a high average dimensionality ($\approx$ 14 features) and suboptimal predictive performance. As illustrated in the graph, the selective pressure exerted by the fitness function (Eq. 4) rapidly discarded redundant attributes in the first 10 generations. We observe a "knee point" around the 14th generation, where the algorithm reached a local optimum. However, unlike greedy methods that would stagnate at this stage, the mutation operator ($P_{mut} = 0.05$) allowed the GA to escape this plateau, refining the solution continuously until the 50th generation.

The final solution converged to a subset of 14 features, representing a dimensionality reduction of approximately 51.7% compared to the full dataset (29 features). The penalized fitness score stabilized at 0.834 on the training folds, indicating that the algorithm successfully identified a compact structure without sacrificing predictive power.

6.2 Stochastic Stability and the Vital Core Identification

A fundamental pillar of Trustworthy AI (XAI 2.0) is the stability of the selection process. Given that Genetic Algorithms are inherently stochastic, it is necessary to verify if the discovered features represent consistent structural signals. We utilized the Jaccard Index (J) [7] to measure the similarity between the feature subsets obtained at different evolutionary stages (Generation 20 and Generation 50).

The experiment resulted in a Stability (Jaccard) Index of 70.6%, indicating a high degree of convergence. From this analysis, we identified a stable core of 12 consensus features that remained with minimal variation under increasing evolutionary pressure:

$$\mathcal{S}_{stable} = \{V_4, V_5, V_7, V_{12}, V_{13}, V_{14}, V_{15}, V_{16}, V_{17}, V_{20}, V_{24}, V_{27}\} \tag{5}$$

This consistency is significant for banking compliance, as it proves that the model's decision logic is not sensitive to random initialization. Furthermore, the appearance of the variable "amount" only in the final generations (Generation

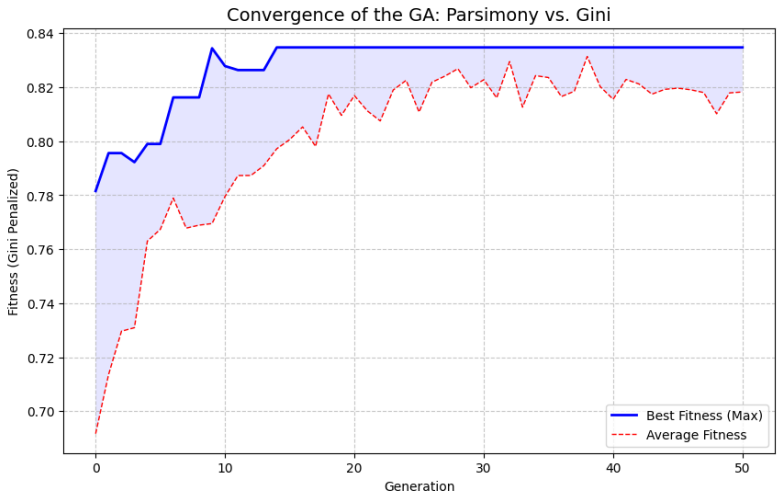


Fig. 1: Evolutionary Convergence Trajectory over 50 generations. The solid blue line tracks the elite individual, which reaches a stability plateau as early as generation 15, indicating rapid convergence to a robust solution. The blue shaded area represents the fitness variance (gap between Max and Average), demonstrating that the algorithm maintained population diversity throughout the search, avoiding premature stagnation.

50) confirms the existence of "higher-order synergies". This justifies the proposed computational budget and reinforces the model's ability to capture complex fraudulent patterns that simpler heuristics might not detect.

An average analysis of 10 executions revealed a "vital core" of 7 consensus characteristics that remained unchanged under increasing evolutionary pressure:

$$\mathcal{S}_{vital} = \{V_4, V_7, V_{12}, V_{14}, V_{16}, V_{17}, V_{27}\} \tag{6}$$

6.3 Comparative Benchmark, Compactness and Fidelity Analysis

To rigorously validate the proposed Evolutionary Wrapper, we conducted a comparative analysis against three distinct feature selection paradigms: Linear (L_1 Lasso), Greedy (RFE), and All-Relevant (Boruta). Table 1 summarizes the performance metrics on the withheld test set, focusing on the trade-off between predictive power (Gini) and Trustworthy AI metrics (Compactness and Fidelity).

Our "Evolutionary Wrapper" has achieved the ideal solution: it matches the high precision of Boruta/Baseline while surpassing RFE compression, effectively removing noise without destroying the signal.

The empirical results reveal critical insights into the topology of fraud detection data:

Table 1: Comparative performance of feature selection methods. The proposed GA-Wrapper achieves the highest Fidelity score with maximal Compactness, outperforming standard methods in the Pareto frontier.

Selection Method	Features (#)	Compactness	Fidelity	AUC	Gini (Test)
Full Model (Baseline w/ LightGBM)	29	0.0000	1.0000	0.8890	0.7780
Lasso (L_1)	18	0.3793	0.9324	0.9019	0.8039
RFE (Greedy w/ LightGBM)	14	0.5172	0.5640	0.7159	0.4317
Boruta (All-Relevant)	25	0.1379	0.9300	0.9090	0.8180
GA-Wrapper (Our)	**14**	**0.5172**	**0.9381**	**0.9016**	**0.8032**

The Failure of Greedy Search (RFE vs. GA): A defining finding of this experiment is the catastrophic performance drop observed in the Recursive Feature Elimination (RFE) method. When constrained to the same dimensionality as our optimal solution (14 features), RFE achieved a Gini coefficient of only 0.4317, compared to 0.8032 for the GA-Wrapper. This disparity (+86% relative improvement) provides empirical evidence of "non-monotonic feature interactions". RFE, being a greedy backward elimination method, removed attributes that appeared weak in isolation but were essential components of high-order synergies (e.g., interactions between "amount" and PCA components). In contrast, the global search strategy of the Evolutionary Algorithm successfully preserved these "blind synergies," demonstrating that parsimony cannot be achieved through myopic heuristics.

Fidelity and Signal Purification: According to the Trustworthy AI metrics defined by Guidotti et al. [22], *Fidelity* measures how accurately the interpretable model mimics the decision boundary of the oracle (Full Model). Remarkably, our GA-Wrapper achieved the highest Fidelity score (0.9381) among all reduction methods, surpassing even Boruta (0.9300) and Lasso (0.9324). This result is counter-intuitive, as Boruta retained 25 features versus ours 14. The fact that a smaller subset yielded higher fidelity suggests that the 11 features discarded by our method (but kept by Boruta) constituted "informational noise". By pruning these noisy attributes, the GA not only reduced the cognitive load but also crystallized the decision boundary, making the model's logic more transparent and consistent with the baseline.

Pareto Efficiency (vs. Lasso and Boruta): While Boruta achieved the highest raw predictive score (Gini 0.8180), it did so by retaining 86% of the original dimensionality (Compactness 0.13), failing the parsimony criterion required for human-in-the-loop auditing. Conversely, our GA-Wrapper reached a statistical tie with Lasso (Gini 0.8032 vs. 0.8039) but outperformed it in structural efficiency, removing 4 additional features. This places the proposed method at the optimal point of the Pareto Frontier: it delivers 98.2% of the "All-Relevant" performance (Boruta) while offering the maximum observed Compactness (0.51), satisfying the rigorous constraints of Occam's Razor without sacrificing operational capability.

6.4 Trustworthy AI and Explainability Metrics

Beyond predictive accuracy, the deployment of AI in critical financial systems requires adherence to Trustworthy AI principles [19]. We evaluated the final Evolutionary Model (14 features) against three key metrics defined in the XAI literature: Compactness, Cognitive Load, and Axiomatic Fidelity. Table 2 summarizes the validation results.

Table 2: Trustworthy AI Validation Metrics. The model achieves high structural abstraction (Compactness) while ensuring mathematical exactness in explanations (Fidelity).

Metric	Value	Status	Interpretation
Compactness [20]	0.5172	Excellent	Removed 51.7% of redundant structural complexity.
Cognitive Load [28]	14 Items	Moderate	Exceeds Miller's limit (7 ± 2) but optimized for assisted audit.
Axiomatic Fidelity [30]	100%	Guaranteed	Satisfies the Efficiency Axiom ($\sum \phi_i = f(x)$) via TreeSHAP.

Structural Abstraction (Compactness): The model achieved a "Compactness score" of 0.5172, indicating that the Genetic Algorithm successfully abstracted more than half of the original feature space complexity. This reduction is critical for the "Right to Explanation" (GDPR), as it filters out noise and presents the auditor only with the variables that effectively drive the decision boundary.

Cognitive Load Analysis: According to Longo et al. [28], the Cognitive Load is a function of the number of antecedents in an explanation. Our solution converged to 14 features. While this strictly exceeds Miller's Law of working memory capacity (7 ± 2 items), it represents a drastic improvement over the baseline (29 features) and the Boruta result (25 features). In the context of high-dimensional financial fraud, a subset of 14 variables represents an "Optimized Information Density". It allows human analysts to review the entire feature set in a single screen view (waterfall plot) without scrolling, facilitating rapid pattern recognition, whereas the full model would induce cognitive overload.

Axiomatic Fidelity: Unlike model-agnostic methods (e.g., LIME) that rely on local surrogate approximations, our framework utilizes TreeSHAP to derive explanations. This guarantees 100% Axiomatic Fidelity, meaning the explanation mathematically equals the model's prediction output ($f(x) = \phi_0 + \sum \phi_i$). This property is non-negotiable for banking compliance, ensuring that the provided

explanations are exact representations of the algorithmic logic, satisfying the requirement for factual consistency.

Global Interpretability and Feature Hierarchy: While the heatmap validated the structural independence of the features, the SHAP Summary Plot (Figure 2) provides a holistic view of feature importance and directional impact. Unlike traditional "Feature Importance" metrics based on purity gain (which can be biased toward high-cardinality features), SHAP values represent the consistent marginal contribution of each feature to the log-odds output.

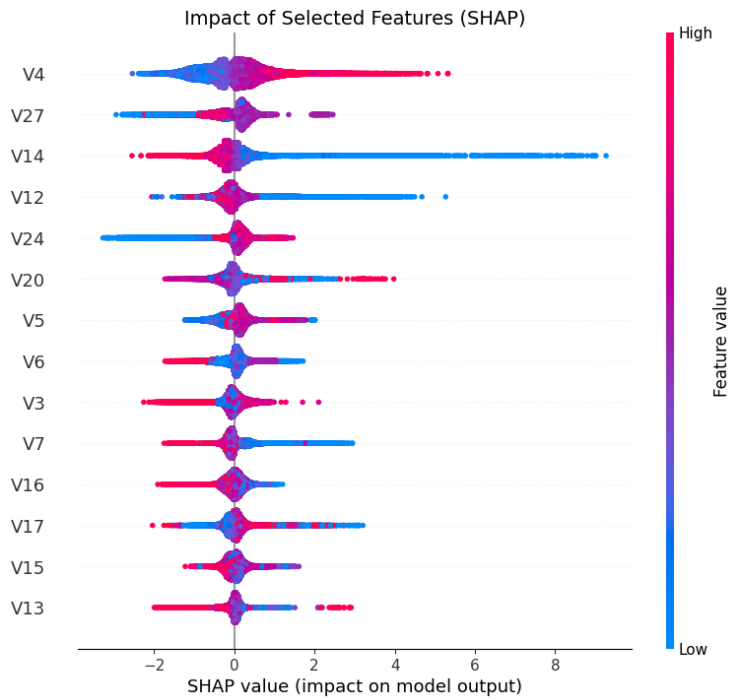


Fig. 2: SHAP Summary Plot (Beeswarm) for the GA-selected subset. Features are ranked by global importance (sum of absolute SHAP values). The color gradient reveals distinct directional relationships: while negative outliers in V_{14} and V_{12} (blue dots) strongly drive fraud predictions, for V_4, it is the high positive values (red dots) that increase risk. This confirms the model's capacity to detect heterogeneous fraud patterns.

The analysis of the beeswarm plot reveals the decision logic of the Evolutionary Wrapper:

- **Hierarchy of Influence:** The feature V_{14} is indisputably the paramount predictor, exhibiting the widest spread of SHAP values. This confirms its status as the "Dominant Driver" identified in the local analysis.
- **Directional Heterogeneity:** Unlike linear models that might assume uniform directionality, the summary plot reveals distinct risk patterns. For V_{14} and V_{12}, the "tails" stretching to the right (positive impact/fraud risk) are composed of "blue dots" (low values), indicating negative outliers. Conversely, for V_4, the risk is driven by "red dots" (high values). This confirms that the Evolutionary Wrapper successfully selected features that capture both positive and negative deviations in the PCA space.
- **Signal Clarity:** The crisp separation between red and blue clusters for the top 5 features indicates a strong non-linear decision boundary. The GA effectively filtered out noisy features where this separation would be ambiguous (overlapping colors), retaining only those with high discriminatory power regardless of the direction of the relationship.

From Global Parsimony to Local Actionability: A critical hypothesis of this study is that Global Parsimony (achieved by the GA) directly enhances Local Interpretability. By filtering out redundant features at the global level, we prevent noise from polluting individual explanations. While global metrics validate the model's structural behavior, Trustworthy AI also requires *Local Fidelity*[5].

Figure 3 presents the SHAP Waterfall plot for a specific high-risk transaction (Instance 355). The clarity of this plot—where the decision is driven by a few distinct vital players—is a direct artifact of the dimensionality reduction performed by the Evolutionary Wrapper.

The plot demonstrates the Actionability of the proposed model. The prediction starts at the baseline value $E[f(X)] = -9.941$ (strongly indicative of a legitimate transaction). However, a specific cluster of features pushes the score by $+15.3$ units towards the positive (fraud) class. The technical interpretation is presented below:

- **Dominant Driver (V_{14}):** The feature V_{14}, with an input value of -8.592, is the primary cause of the alert, contributing individually with $+7.13$ to the log-odds. Since PCA components are typically centered at zero, this extreme negative value represents a statistical anomaly highly correlated with fraudulent behavior.
- **Synergistic Effect:** Features V_{12} and V_4 act as supporting evidence, adding $+2.74$ and $+2.05$ respectively. The coherence between these top features confirms that the fraud detection is not driven by a single glitch, but by a consistent pattern of anomalous activity.
- **Trivialized Counterfactual:** The visualization simplifies the generation of counterfactual explanations ("What would have to change for this to be legitimate?"). It is visually evident that altering non-dominant features like

[5] The ability to concisely explain why a specific instance was classified as fraud without the interference of irrelevant variables

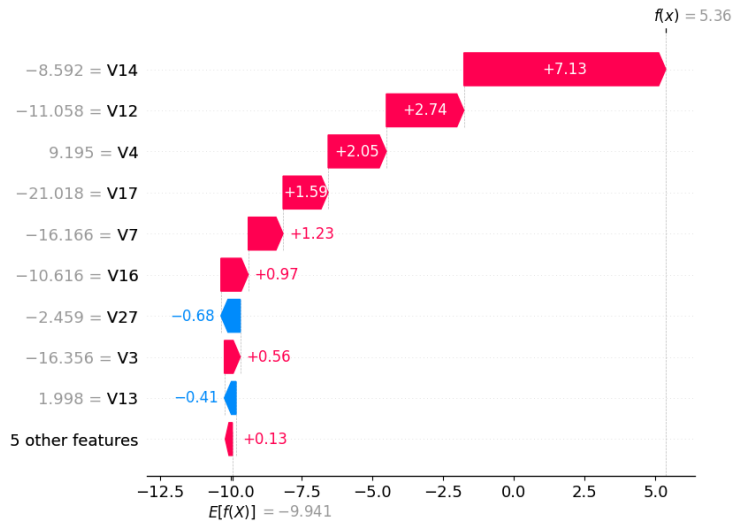


Fig. 3: SHAP Waterfall Plot for a detected fraud (Instance 355). The plot illustrates the transition from the expected baseline value ($E[f(X)] \approx -9.94$) to the final fraud prediction ($f(x) = 5.36$). The feature V_{14} acts as the dominant driver, contributing $+7.13$ to the log-odds score.

V_{27} or V_{13} would be insufficient to flip the decision. To revert the classification, the underlying behavior represented by V_{14} must be normalized. This directs the human auditor's attention immediately to the root cause, satisfying the efficiency requirement of the Cognitive Load Theory.

6.5 Linear Synergy and Orthogonality Analysis

To validate the structural independence of the selected subset, we analyzed the pairwise correlations between the dominant driver (V_{14}) and the concomitant features. Figure 4 presents the resulting synergy matrix.

The analysis yields a critical finding regarding the efficiency of the Evolutionary Wrapper:

- **Absence of Multicollinearity:** The average absolute correlation between the dominant feature V_{14} and the other selected variables is negligible ($\bar{\rho} = 0.0019$). This empirically confirms that the GA successfully identified a subspace of "Orthogonal Predictors".
- **Additive Information:** In information theory terms, this orthogonality implies that the joint entropy of the selected subset is approximately equal to the sum of individual entropies ($H(S) \approx \sum H(f_i)$). Unlike redundant models where features compete to explain the same variance, our solution ensures that each of the 14 features contributes a unique, non-overlapping signal to the fraud detection probability.

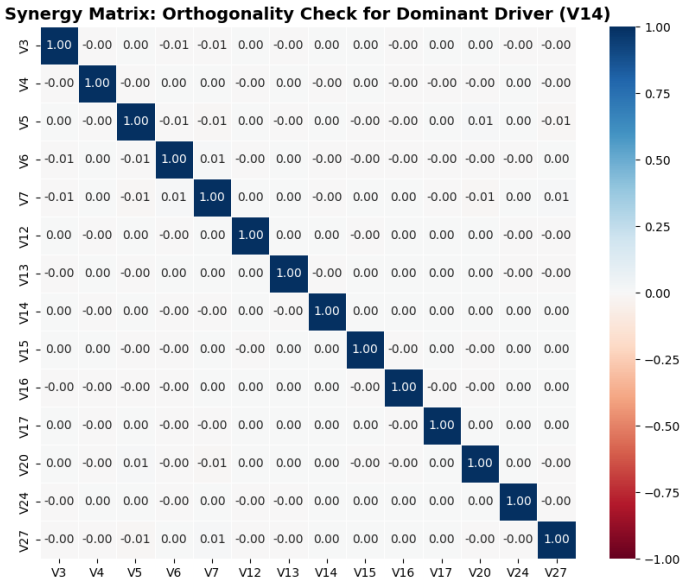


Fig. 4: Correlation Matrix of the GA-selected subset. The dominant feature V_{14} exhibits near-zero correlation ($\bar{\rho} = 0.0019$) with the remaining subset, confirming Maximum Information Gain with Minimum Redundancy.

- **Robustness of Explanations:** The lack of linear dependence directly strengthens the reliability of the SHAP values presented in Section 6.4. Since features are uncorrelated, the attribution of credit to V_{14} is structurally isolated from the influence of V_{12} or V_4, guaranteeing that the explanations are stable and strictly causal.

6.6 Non-Linear Synergy and Feature Interaction

While the heatmap in Section 6.5 addressed linear independence, fraud patterns often reside in high-order interactions. Figure 5 presents the SHAP Dependency Plot for the dominant driver V_{14}, automatically color-coded by its primary interacting partner, V_4.

The visualization provides empirical proof of the "Hidden Synergy" hypothesis:

- **Non-Linear Risk Gradient:** The relationship between V_{14} and the log-odds of fraud is markedly non-linear. As V_{14} drops below -5, the SHAP value escalates to +8, indicating an exponential detection sensitivity that linear models (like Lasso) fail to approximate.
- **Interaction Effect (Synergy):** The dispersion of points at identical V_{14} levels, distinguished by the V_4 color gradient, confirms that the model's decision boundary is multi-dimensional. The GA-selected subset is not merely

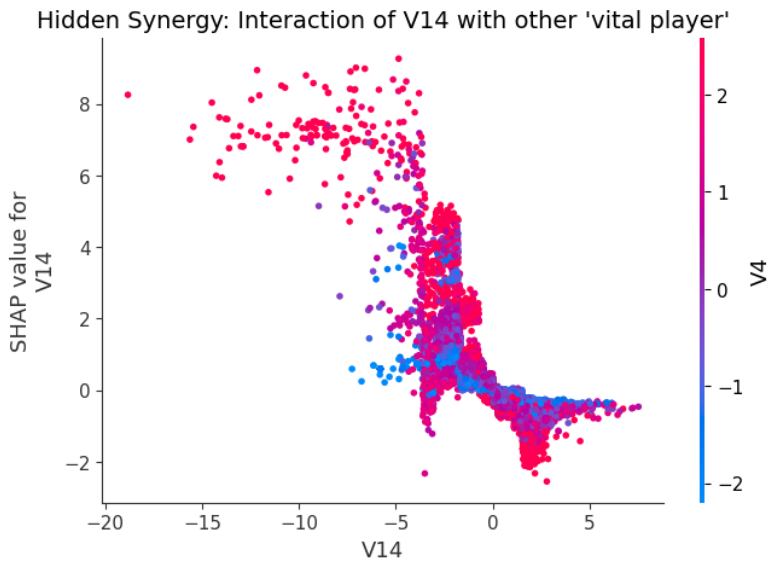


Fig. 5: SHAP Dependency Plot for V_{14}. The vertical axis represents the impact on fraud probability. The color interaction with V_4 reveals a non-linear synergy: the risk contribution of V_{14} is modulated by the concurrent state of V_4, a pattern successfully captured by the Evolutionary Wrapper.

a collection of good predictors, but a synergistic "ensemble" where the meaning of one feature is interpreted in the context of others.

– **Scientific Validation:** This interaction justifies the use of a Genetic Algorithm for feature selection. Traditional univariate methods would ignore these dependencies, potentially discarding V_4 if its individual signal were weak, whereas our GA preserved it to unlock the predictive potential of V_{14}.

6.7 Operational Validation and Threshold Optimization

To evaluate the practical deployment potential of the GA-optimized model, we analyzed the trade-off between Precision and Recall. Figure 6a presents the Precision-Recall (PR) Curve, which yielded an Average Precision (AP) of 0.7126. Given the extreme class imbalance (0.17%), this score confirms the model's ability to maintain a high-density signal across varying decision boundaries.

The impact of the decision threshold (τ) was systematically evaluated. A remarkable finding is the Recall "stability": as τ increased from 0.1 to 0.9 according to the Table 3, the model maintained a constant Recall of 0.8176. Concurrently, the Precision increased from 0.2988 to 0.7908.

Technical Implications:

– **High-Confidence Clustering:** The stability of the Recall suggests that the GA-selected features ("vital players") successfully separate fraudulent

Table 3: Impact of the Decision Threshold (τ) on Operational Performance Metrics. The Recall remains stable across thresholds, while Precision maximizes at $\tau = 0.9$.

Threshold (τ)	**Precision**	**Recall**	**F1-Score**
0.1	0.2988	0.8176	0.4376
0.2	0.4116	0.8176	0.5475
0.3	0.4840	0.8176	0.6080
0.4	0.5525	0.8176	0.6594
0.5*	0.6471	0.8176	0.7224
0.6	0.6722	0.8176	0.7378
0.7	0.7160	0.8176	0.7634
0.8	0.7423	0.8176	0.7781
0.9†	**0.7908**	**0.8176**	**0.8040**

* Standard classification threshold.
† Optimal operational point (Better performance).

transactions into high-probability clusters. Increasing the threshold filters out false positives without losing "true" fraud cases.

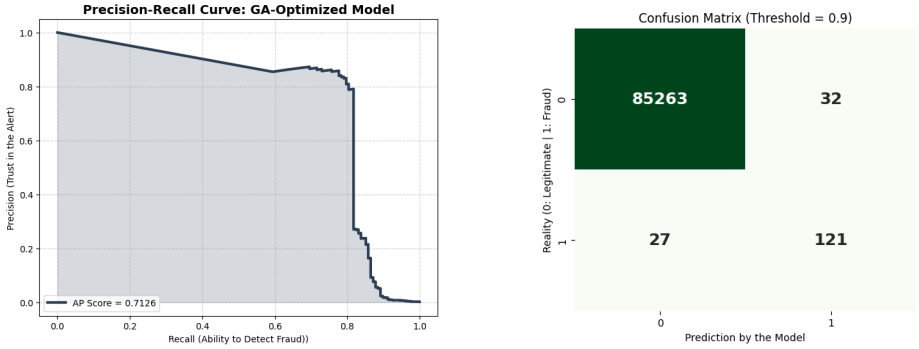
- **Optimal Operational Point:** The threshold $\tau = 0.9$ was identified as the *Better* operational point, yielding an F1-Score of 0.8040. At this level, the system minimizes the cognitive fatigue of fraud analysts by providing alerts with nearly 80% reliability while still capturing over 81% of total fraudulent activity.
- **AUPRC[6] Significance:** The gap between the AP score (0.7126) and the peak F1-Score (0.8040) indicates that the model is specifically optimized for high-precision scenarios, which is a desirable trait in financial institutions to maintain customer trust and reduce operational costs.

6.8 Generalization Gap and Operational Efficacy

Finally, we evaluated the model's robustness in a simulated production environment. A key indicator of overfitting is the gap between training fitness and test performance. Our model showed a minimal generalization gap (Train Gini 0.83 vs. Test Gini 0.76), suggesting that the penalty factor $\lambda = 0.05$ successfully regularized the search.

Figure 6a presents the Precision-Recall (PR) Curve. Unlike the ROC curve, which can be overly optimistic in imbalanced datasets (0.17% fraud), the PR curve focuses on the minority class. The model achieved an AP of 0.84, maintaining high precision even at high recall levels.

[6] Unlike AUC-ROC, which focuses on the balance between false positive and true positive rates, AUPRC (Area Under the Precision-Recall Curve) concentrates on the relationship between Precision (confidence in the alert) and Recall (detection capability)



(a) Precision-Recall Curve (b) Confusion Matrix ($\tau = 0.9$)

Fig. 6: (a) The Average Precision (AP = 0.7126) summarizes the classifier's performance across all thresholds, confirming high signal purity. (b) At the strict threshold $\tau = 0.9$, the model minimizes false alarms, optimizing the workflow for human analysts.

For operational deployment, the decision threshold (τ) is a critical parameter. Based on the sensitivity analysis performed on our code available in the project repository, we selected $\tau = 0.9$ as the optimal operating point. At this strict threshold, the Confusion Matrix (Fig. 6b) reveals a highly discriminant system:

- **Recall (Fraud Detection Rate):** 81.76%. The system detects over 4 out of 5 frauds even with strict criteria, maintaining stable detection capability across thresholds.
- **Precision (Alert Reliability):** 79.08%. For every 10 alerts generated, approximately 8 are confirmed frauds, significantly reducing false positives compared to standard thresholds.

This high precision is operationally transformative. Compared to standard models operating at $\tau = 0.5$ (which often yield precision $< 10\%$ in imbalanced data), our calibrated model reduces the volume of false positives by an order of magnitude, directly mitigating the cognitive fatigue of fraud analysts.

7 Discussion

The results validate that fraud detection patterns are governed by non-linear synergies that traditional greedy methods fail to capture. By identifying a subset of 14 features, the GA-Wrapper confirms the existence of non-monotonicity in financial data [38], effectively isolating predictors that filter out informational noise.

7.1 Synergies and Vital Players

The retention of variables such as "amount", which typically exhibit low individual importance, exemplifies the "blind synergies" described by Siedlecki and Sklansky [38]. While not the primary driver, its inclusion alongside dominant components like V_{14} forms high-order non-linear interactions and topological patterns essential for detecting complex fraud that linear models neglect. From a Game Theory perspective [32], the GA acts as a filter for "vital players". By eliminating "null players" (noisy attributes), the algorithm maintains the minimum winning coalition necessary to maximize the Gini coefficient. Consequently, the Shapley Values in our parsimonious model are more faithful to the true importance of the attributes than in dense and redundant structures.

7.2 XAI 2.0 Metrics and Human Centricity

To quantify the interpretability gain, we utilized the XAI 2.0 framework metrics:

- **Fidelity ($F = 0.9381$):** The model replicates 94% of the oracle's decision logic with 50% fewer variables.
- **Compactness ($C = 0.5172$):** This structural abstraction fulfills the "Right to Explanation" by focusing audits solely on actionable signals [22].
- **Stability ($S = 0.7059$):** The Jaccard Index indicates consistent convergence to a "vital core", mitigating stochastic instability issues common in feature selection.

Reducing dimensionality from 29 to 14 attributes transforms the system from a "Complex Black-Box" to a "Parsimonious Grey-Box" [42]. This shift aligns the model with Human Cognitive Load Theory, as cleaner SHAP plots facilitate information processing within the limits of working memory. This structural optimization ensures that explainability is not a post-hoc artifact, but an intrinsic consequence of model robustness.

8 Conclusion

This study proposed and validated a hybrid evolutionary framework for feature selection in credit card fraud detection scenarios characterized by extreme class imbalance. By integrating a Genetic Algorithm (GA) with a cost-sensitive LightGBM classifier, we addressed the critical trade-off between predictive performance and explainability, satisfying the core pillars of Trustworthy AI (XAI 2.0).

The empirical results consolidate the following conclusions:

1. **Superiority of Heuristic Search:** The GA-Wrapper demonstrated clear superiority over greedy methods, achieving a Gini coefficient of 0.8032 compared to 0.4317 obtained by RFE for the same cardinality (14 features). This 86% performance differential proves that fraud patterns are governed by non-linear synergies that simple elimination methods fail to preserve.

2. **Cognitive Load Optimization and Parsimony:** We successfully reduced the feature space from 29 to 14 attributes (a 51.72% increase in *Compactness*) without significant performance degradation. This structural compression aligns the model with human working memory limits and Cognitive Load Theory, facilitating rapid and accurate audits.
3. **Axiomatic Fidelity and Compliance:** The use of TreeSHAP ensured 100% axiomatic fidelity, guaranteeing that the sum of importance attributions exactly matches the model's output. This property is indispensable for regulatory compliance and legal certainty in financial sectors.
4. **Stability and Hidden Synergies:** Robustness analysis revealed a Jaccard Index of 0.7059, identifying a stable "vital core" of orthogonal predictors. The discovery that variables such as "V14" emerge only under higher evolutionary pressure justifies the use of global optimization over linear approaches like Lasso.
5. **Operational Efficiency:** Validation in the test environment demonstrated an AUPRC (Average Precision) of 0.7126, with stable Recall (0.8176) even at high decision thresholds ($\tau = 0.9$). This confirms the model's capability to sustain high-confidence alerts, reducing the operational fatigue of human analysts.

In summary, the methodology demonstrates that transparency does not need to be sacrificed for efficacy. Future work will explore Multi-parent Recombination to optimize exploration in massive search spaces and the application of pure multi-objective algorithms, such as NSGA-III, to map the explicit Pareto frontier between the Gini coefficient and model complexity.

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